The University of the South Pacific

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School of Computing, Information and Mathematical Sciences

CS211: Computer Organisation

Test 2 – Semester 1, 2014

Time Allowed: 1 hour

Total Marks: 40 (15% of final grade)

Solution.

Name: _____ **Student ID:** _____

INSTRUCTIONS:

- There are 9 questions in this paper and all are compulsory.
- All answers must be written in the spaces provided in this booklet.
- You are allowed to use a silent, non-programmable calculator.
- Show all working.

Question 1

For the truth table given below:

x	y	z	F(x,y,z)
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

a) Write the disjunctive normal form for the above truth table. (2 marks)

(2) $f(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z$

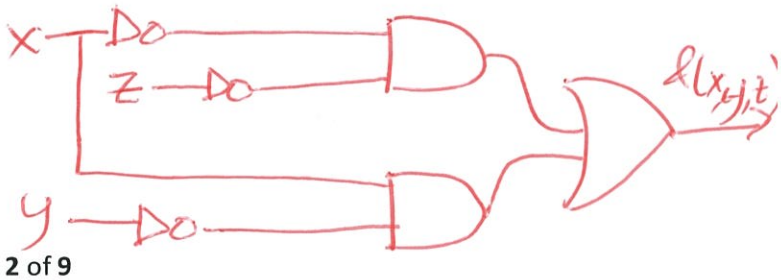
b) Simplify the above function using Boolean algebra and show an implementation using logic gates. (4 marks)

$\bar{x}\bar{y}\bar{z} + \bar{x}y\bar{z} + x\bar{y}\bar{z} + x\bar{y}z$

(2) $\bar{x}\bar{z}\cancel{(\bar{y}+y)} + x\bar{y}\cancel{(\bar{z}+z)}$

$\bar{x}\bar{z} + x\bar{y}$

(2)



Question 2

Assume you have the following page reference stream: A, B, C, D, A, B, E, A, B, C, D, E.

a) Assuming a page cache of size 4 pages and a FIFO replacement policy, how many misses will there be? (2 marks)

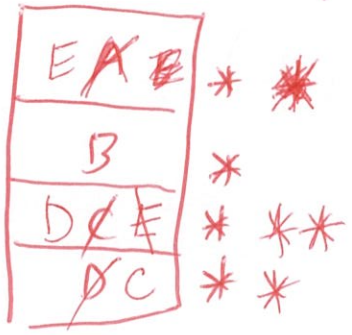
(1)

10 misses (1).

b) Assuming a page cache size of size 4 pages and LRU replacement policy, how many misses will there be? (2 marks)

A B C D A B E A B C D E

(1)

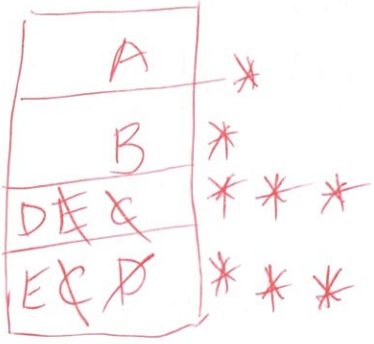


8 misses (1).

c) Assuming a page cache size of size 4 pages and LFU replacement policy, how many misses will there be? (2 marks)

A B C D A B E A B C D E

(1)



8 misses (1)

Question 3

A segmented address space uses paged virtual memory. Each virtual address has a 2 bit segment number, a 2 bit page number and an 8 bit offset within the page. In the following page tables, a "-" indicates that the page is not in memory, which has 8 page frames.

segment 0		segment 1		segment 2		segment 3	
page	frame	page	frame	page	frame	page	frame
0	-	0	-	0	-	0	-
1	1	1	5	1	3	1	4
2	-	2	6	2	2	2	-
3	0	3	-	3	-	3	7

a) What is the lowest hexadecimal virtual address in segment 2 that will cause a page fault? (2 marks)

10 00 | 0000 | 0000 (1)

800_{16} (1)

b) Find the real memory address (in hexadecimal) if the virtual hexadecimal address is F12. (The result is "*page fault*" if the page is not in real memory.) (3 marks)

F12

(1) VA = 1111 00010010
seg Pg Offset

(1) Seg 3, Pg 3 ⇒ frame 7

PA = 111/0001/0010 ⇒ 712 (1)

c) Find the virtual address (in decimal) if the real memory decimal address is 554. (3 marks)

554 = 10 00101010 → Frame 2, Segment 2 Page 2

⇒ 10 10 00101010 2602

Question 4

Justify whether it is possible to design an expanding opcode to allow the following to be encoded in a 12-bit instruction? A register is 3 bits. (4 marks)

6 instructions with three registers
250 instructions with one register
8 instructions with zero registers

000 A B C
⋮ ⋮ ⋮ ⋮ } 6 (1)
101 A B C

110 000000 C
⋮ ⋮ ⋮ } 64 (1)
110 111111 C

111 000000 C
⋮ ⋮ } 64 (1)
111 111111 C

128

∴ This design is not possible. only 128 instructions out of 250 instructions were possible with one register. (1)

Question 5

Assume the following instructions are available for 0-address (*stack*-based) and 2-address machines (*Register* Machine): ADD, SUB, MUL, DIV, MOVE, LOAD, STORE, PUSH, POP. Memory addresses like A, B, C, etc. are used as operands. Using these instructions, show how to compute the expression $X = (A \times B) - C \times D$ using:

a) 0-address machine. (2 marks)

Push A
Push B
mul
Push C
Push D
mul
Sub
Pop X

b) 2-address machine (assume an instruction like "*SUB R0, Q*" puts *R0 – Q* into *R0*). (2 marks)

move R0 A
mul R0 B
move R1 C
mul R1 D
Sub R0 R1
move X R0

Question 6

Consider a virtual address space of eight pages of 1024 words each mapped, onto a physical memory of 32 frames.

a) How many bits are there in virtual address? (2 Marks)

Logical Address = 3 + 10 = 13 bits.

b) How many bits are there in physical address? (2 Marks)

Physical Address = 5 + 10 = 15 bits.

Question 7

Assume you have a computer which currently does not have any cache memory. Main memory access time is 150 *nsec* and disk access time is 2500 *nsec*. The memory hit ratio is 60%. A faster memory is always accessed before a slower one. You want to add a cache that has access time of 90 *nsec* such that the average access time for your computer becomes 365 *nsec*. What should you set as the cache hit ratio to achieve this average access time? (3 marks)

(1) let p = cache hit ratio

Avg Access time = cache + memory + disk.

(1) $365 \Rightarrow p90 + 0.6(150+90) + (0.4-p)(2500+150+90)$

$p = 0.3302$.

(1) $p \Rightarrow 33.02\%$.

Question 8

Suppose you design an instruction that takes 8 *nsec* to execute. You need to buy a clock that can allow each instruction to complete within one clock cycle. Would you buy a 200 MHz clock? Why or why not? (2 marks)

200 MHz → 200×10^6 cycles/sec (1)

1 cycle = 5 nsec

5 nsec < 8 nsec

∴ No I will not buy as it will not complete withone cycle

Question 9

Convert the following formula from infix to reverse polish notation. (3 marks)

$$(A + B)/C + E \times F/((G + H) \times I)$$

1. AB+C/ EFx GH+Ix/+

2. ~~AB+ C/ EFGH+IX~~

AB+ C/ EF GH+Ix/x+

The End

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